

CSA1006 – SOFTWARE ENGINEERING

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UNIT 1 – ASSESSMENT

Problem 5: Ethical Evaluation of an AI Hiring System

Scenario

A software company is developing an **AI-based hiring system** that automatically screens resumes, ranks candidates, and recommends applicants for interviews. During testing, the team discovers that the AI model unfairly rejects applicants from certain demographic groups because the training data contains historical hiring bias.

To ensure fairness, privacy, and responsible AI development, the company decides to redesign the system using ethical AI principles.

1. Ethical Issue Identification

The AI hiring system faces several ethical challenges:

a) Algorithmic Bias

The AI favors certain applicants while rejecting equally qualified candidates from specific gender, age, or ethnic groups due to biased training data.

b) Lack of Transparency

Applicants do not know why they were rejected because the AI decision-making process is not clearly explained.

c) Unfair Decision Making

Qualified candidates may lose employment opportunities because of inaccurate or discriminatory AI predictions.

d) Accountability

If the AI makes an incorrect hiring decision, it becomes difficult to determine whether the responsibility lies with the developers, HR team, or organization.

e) Loss of Human Judgment

Completely relying on AI may ignore important qualities such as creativity, leadership, and communication skills that humans can better evaluate.

Example:

If the AI was trained using historical company data where mostly male candidates were hired, it may incorrectly rank female applicants lower even when they have the same qualifications.

2. Actions Based on Data Ethics Principles

To develop a responsible AI hiring system, the company should follow these data ethics principles:

1. Fairness

Train the AI using diverse and balanced datasets to reduce discrimination against any applicant group.

2. Transparency

Provide applicants with understandable explanations for hiring decisions and the factors influencing the AI recommendations.

3. Accountability

Maintain human oversight so HR managers review AI recommendations before making final hiring decisions.

4. Privacy

Collect only the information necessary for recruitment and securely protect applicant data.

5. Reliability

Continuously test, monitor, and improve the AI model to ensure accurate and unbiased performance.

3. Potential Privacy Risks

The AI hiring system may introduce the following privacy risks:

a) Personal Data Leakage

Unauthorized access may expose applicant names, addresses, phone numbers, educational records, or resumes.

b) Data Misuse

Collected applicant information may be used for purposes beyond recruitment without proper consent.

c) Unauthorized Access

Weak security controls may allow hackers or unauthorized employees to access confidential hiring data.

d) Data Breach

Sensitive candidate information may be stolen through cyberattacks.

e) Excessive Data Collection

Collecting unnecessary personal information increases privacy risks and violates data minimization principles.

4. Recommended Ethical Design Measures

To make the AI hiring system secure and ethical, the following practices should be implemented:

1. Bias Detection and Fairness Testing

Regularly evaluate AI predictions to identify and remove discriminatory behavior.

2. Human-in-the-Loop Decision Making

Allow HR professionals to review and approve AI recommendations before final hiring decisions.

3. Explainable AI (XAI)

Provide clear explanations showing why each candidate received a particular score or recommendation.

4. Data Encryption

Encrypt applicant information during storage and transmission to prevent unauthorized access.

5. Access Control

Allow only authorized HR personnel to access sensitive recruitment data.

6. Regular Security Audits

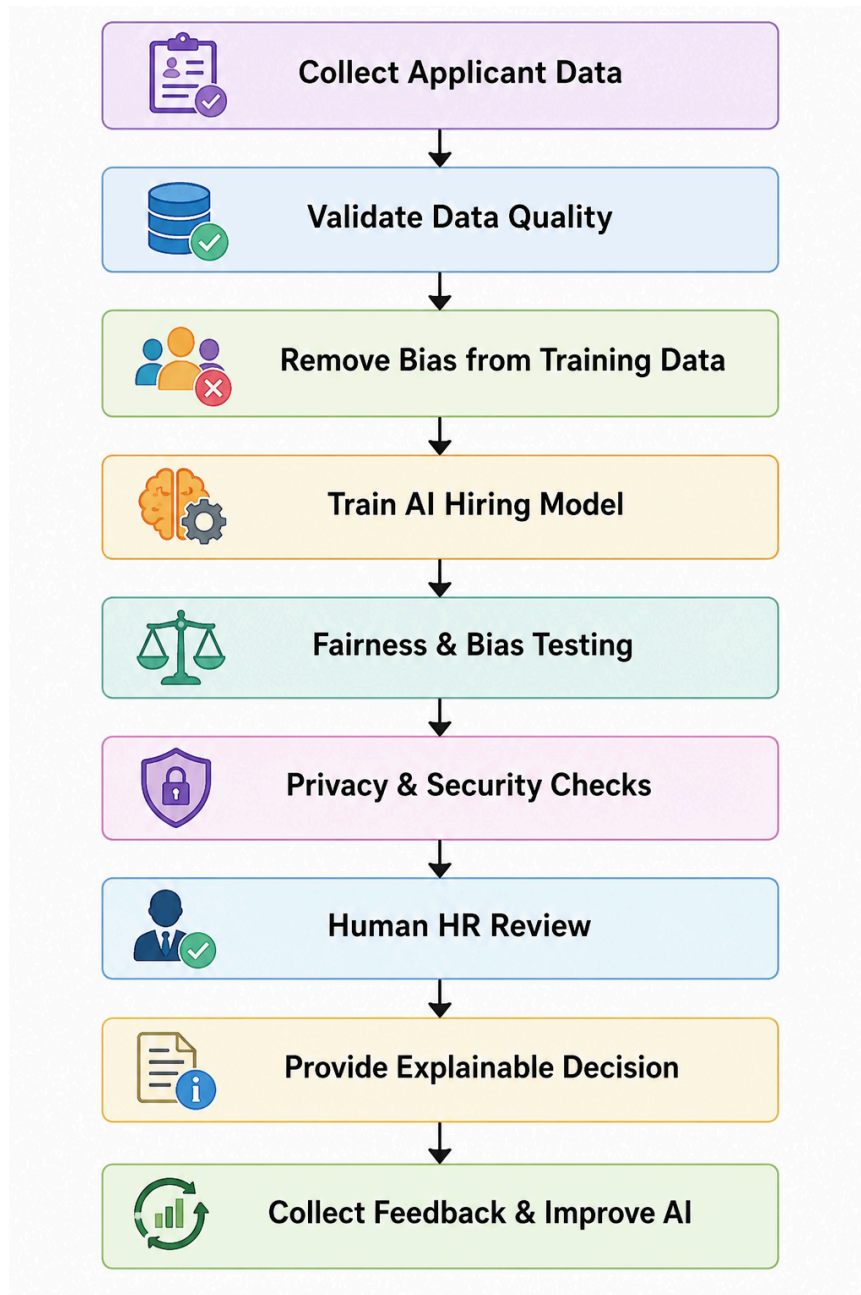
Conduct periodic security testing and compliance reviews to identify vulnerabilities.

7. Data Minimization

Collect only information necessary for evaluating applicants and securely delete unused data.

5. Ethical AI Development Workflow

Workflow Description



Explanation

The hiring process begins by collecting only relevant applicant information. The dataset is checked for fairness and bias before training the AI model. After development, fairness testing and privacy checks ensure ethical performance. Human HR managers review AI recommendations before final hiring decisions. Feedback is continuously collected to improve fairness, transparency, and system accuracy.

User Story

As an HR manager, I want the AI hiring system to evaluate applicants fairly and explain its recommendations so that qualified candidates are selected without discrimination.

Acceptance Criteria

- The AI should evaluate candidates without gender, age, or ethnic bias.
- Applicant data must remain confidential and securely stored.
- HR managers should review AI recommendations before final hiring decisions.
- The system should provide explanations for every recommendation.
- The AI model should undergo regular fairness and security testing.
- Applicant data should be collected only with proper consent.

Conclusion

The AI hiring system must balance **fairness, privacy, transparency, and security** to ensure responsible recruitment. By following data ethics principles, reducing bias, protecting applicant information, and maintaining human oversight, the organization can build a trustworthy AI system that supports ethical hiring decisions while improving recruitment efficiency.